Data Analyst Power BI Report

Author : Gouru Rohith Chandu

Task_id : TW-TSK-DA-25-01

Objective: Analyze sales, profit, discounts, returns, customer retention, and product/category

---

1) Sales & Operational Profitability by Region & Segment

1. Added a **Sales & Profitability page** with KPIs (Total Sales, Operational Profit, Profit Margin %, Return Rate %).
2. Created a **Matrix** → Rows = Region + Customer Segment; Columns = Year; Values = Sales, Profit, Margin.
3. Applied **conditional formatting** on Profit (negative = red, positive = green).
4. Built **Top 5 Products by Profit** (bar chart with Top N filter).
5. Added a **Loss-Making Table** → Region + Product Subcategory with negative profit highlighted.
6. Used **Slicers** (Year, Region, Segment, Product Category) for interactive filtering.
7. Insight: Some Regions had high sales but low margins due to discounts/returns.

---

2) Discount vs Profitability (Correlation)

1. Built measures for **Discount_Profit_Correlation** and **DeliveryTime_Profit_Correlation** using custom DAX (Pearson correlation).
2. Displayed correlation results on **Card visuals** (values between -1 and +1).
3. Created a **Table** → Product Category, Net Sales, Avg Discount %, Profit.
4. Applied **conditional formatting** → high discount but low profit categories flagged.
5. Added a measure: **Discount_Hurts_Flag** (1 = high discount & low margin).
6. Built a **Scatter chart** → X = Discount %, Y = Profit, with trend line.
7. Insight: High discounting boosted sales volume but often reduced profitability.

---

3) Returns & Customer Retention Risk

1. Added a **Returns page** with KPI cards → Return Rate %, Repeat Customer Rate, Orders per Customer.
2. Built **Bar chart** → Product Subcategory vs Return Count (sorted).
3. Added **Table** → Subcategory, Return Reason, Return Count, Profit.
4. Created **Pie chart** → Return reasons (Quality, Size mismatch, Delivery).

5. Measured **Repeat Customer Rate** using DAX (customers with >1 orders).
6. Analyzed **Return Rate by Segment & Region** (column chart).
7. Insight: Footwear had high returns due to sizing issues; some regions faced delivery-related returns.

---

4) Drillthrough & Deep Analysis

1. Enabled **Drillthrough pages** → From Category → Product → Order.
2. Drillthrough filters allow moving from summary KPIs to detailed transactions.
3. On **Product Drillthrough page**, showed sales, profit, returns at product-level.
4. On **Order Drillthrough page**, listed order details, discount applied, return flag.
5. Added **Back button** to return to summary page.
6. Insight: Easy navigation from high-level KPIs down to problematic orders.
7. Helps managers track down specific issues quickly.

---

5) Product & Category Operational Deep-Dive (Pareto 80/20)

1. Built ProductProfit_Summary in Power Query → grouped ProductID with Total Sales, Costs, Profit.
2. Added **Running Total** and **Cumulative %** columns in Power Query.
3. Created **Combo chart** → Columns = Profit by Product, Line = Cumulative %.
4. Used **Secondary Axis** for Cumulative % and added 80% **constant line**.
5. Applied **Top N filter** (e.g., Top 50 products) for readability.
6. Added **Year/Quarter/Month slicers** from Date table for time analysis.
7. Insight: ~20% of products contribute ~80% of total profit → focus resources on these.

Final KPIs Tracked

- **Sales & Profitability:** Net Sales, Profit, Profit Margin %, Return Rate %
- **Discount Impact:** Average Discount %, Correlation with Profit, Revenue lost due to discounting
- **Returns & Retention:** Return Rate %, Top Products by Returns, Repeat Customer Rate
- **Drillthrough:** Region → Category → Product → Order level analysis
- **Pareto Analysis:** Profit Contribution %, Total Profit by Category, Revenue per Product